SON 2378: An Integrated Toolkit for Linking Dynamic Water Quality Models to Ecological Assessment Tools

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Problem:

- USACE planners rely on static or simplified environmental inputs to quantify restoration benefits using Habitat Evaluation Procedures and Habitat Suitability Index models.
- Sophisticated, dynamic process-based water quality models exist, but there is no standard method to connect their high-resolution outputs to the ecological models that guide million-dollar investments.
- The result is reduced scientific defensibility, increased project risk, and a potential failure to maximize ecological outcomes.

Envisioned Product:

A Standardized Decision-Support Toolkit

1. A Python-based toolkit that post-processes CE-QUAL-W2 and ClearWater-Riverine outputs into HSI-compatible summary statistics
2. A practitioner's guide with step-by-step workflows and examples for linking WQ and ecological models in common restoration scenarios
3. A demonstration/training package featuring at least two case studies to ensure rapid technology transfer and adoption

Value Added:

- Enhance Scientific Defensibility: Replace qualitative assumptions with validated, dynamic model data.
- Increase Planning Efficiency: Replaces ad hoc, time-consuming data manipulation with a standardized workflow.
- Enable Rigorous Evaluation: Allow planners to assess project alternatives under varying conditions, including future hydrologic scenarios and extreme events.
- Maximize Ecological Return: Ensure that federal restoration investments are directed toward the most effective, risk-informed, and scientifically sound projects.

Regional/National Significance:

- Enterprise-Wide Applicability: Directly supports all USACE Districts by leveraging models already in widespread use through a single standardized methodology.
- Supports Diverse Regional Needs: From salmonid habitat assessments in the Pacific Northwest to freshwater mussel conservation in the Southeast.
- Supports Critical National Priorities: Directly applicable to Harmful Algal Bloom mitigation, Endangered Species Act compliance, and resilience planning. Provides the quantitative basis for more defensible benefit-cost analyses in planning studies nationwide.